

**SIXTH SEMESTER (CUCBCSS—UG) DEGREE [SPECIAL] EXAMINATION
MARCH 2021**

Mathematics

MAT 6B 11—NUMERICAL METHODS

(Multiple Choice Questions for SDE Candidates)

Time : 15 Minutes

Total No. of Questions : 20

Maximum : 20 Marks

INSTRUCTIONS TO THE CANDIDATE

1. This Question Paper carries Multiple Choice Questions from 1 to 20.
2. The candidate should check that the question paper supplied to him/her contains all the 20 questions in serial order.
3. Each question is provided with choices (A), (B) and (C) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

MAT 6B 11—NUMERICAL METHODS

(Multiple Choice Questions for SDE Candidates)

1. Approximate value = True value + _____.
(A) Expected value. (B) Error.
(C) Absolute value.
2. _____ is a root of the equation $x^2 + 4x + 4 = 0$.
(A) 2. (B) -2.
(C) 3.
3. A real root of the equation $x^2 - 2x - 3 = 0$ lies in _____.
(A) $[2, 4]$. (B) $[5, 6]$.
(C) $[1, 2]$.
4. $E f(x) =$ _____.
(A) $f(x + h)$. (B) $f(x - h)$.
(C) $f(x)$.
5. $A =$ _____.
(A) $E - 1$. (B) E .
(C) $E + 1$.
6. Divided difference operator is _____.
(A) Bilinear. (B) Asymmetrical.
(C) Linear.
7. Every square _____ matrix will have an inverse.
(A) Non-singular. (B) Singular.
(C) Upper Triangular.

8. A is an $n \times n$ matrix. Then $AA^{-1} = \underline{\hspace{2cm}}$.
- (A) A. (B) I.
(C) A^2 .
9. In Simpson's $1/3$, n should be $\underline{\hspace{2cm}}$.
- (A) Odd. (B) Even.
(C) Decimal Number.
10. In each step in the Gauss elimination method, the co-efficient of the first unknown in the first equation is called $\underline{\hspace{2cm}}$.
- (A) Term. (B) Pivotal co-efficient.
(C) Pivoting.
11. The Gauss elimination method fails if any one of the pivotal co-efficients becomes $\underline{\hspace{2cm}}$.
- (A) 1. (B) Constant.
(C) Zero.
12. The characteristics polynomial of an $n \times n$ matrix A is $\underline{\hspace{2cm}}$.
- (A) $A - \lambda I$. (B) $|A - \lambda I| = 0$.
(C) $|A - \lambda I|$.
13. $|A - \lambda I| = 0$ of an $n \times n$ matrix A is called $\underline{\hspace{2cm}}$.
- (A) Characteristic polynomial. (B) Characteristic equation.
(C) Eigen Vector.
14. The $\underline{\hspace{2cm}}$ of the characteristic equation $|A - \lambda I| = 0$ is called characteristic roots.
- (A) Co-efficients. (B) Roots.
(C) Degree.
15. If λ is an eigen value, then column vector X such that $\underline{\hspace{2cm}}$ is called an eigen vector associated with the eigen value λ .
- (A) $AX > \lambda X$. (B) $AX < \lambda X$.
(C) $AX = \lambda X$.

Turn over

16. $y' = x^2 + y^2$, then $y''' =$ _____.

(A) $2x + 2yy'$.

(B) $2 + 2yy'' + 2(y')^2$.

(C) $2 + 2yy''$.

17. $y^{(n)} = y_0 + \int_{x_0}^x f(x, y^{(n-1)}) dx, n = 1, 2, 3, \dots$ is the iterative formula for _____.

(A) Euler method.

(B) Picard's method.

(C) Runge-Kutta method.

18. $y_{n+1} = y_n + h f(x_n, y_n), n = 0, 1, \dots$ is the iterative formula for _____.

(A) Picard's method.

(B) Euler method.

(C) Runge-Kutta method.

19. $E^2 f(x) =$ _____.

(A) $f(x)$.

(B) $f(x + 2h)$.

(C) $f(x - 2h)$.

20. $D^2 f(x) =$ _____.

(A) $f(x)$.

(B) $f''(x)$.

(C) $f'(x)$.

**SIXTH SEMESTER (CUCBCSS—UG) DEGREE [SPECIAL] EXAMINATION
MARCH 2021**

Mathematics

MAT 6B 11—NUMERICAL METHODS

Time : Three Hours

Maximum : 120 Marks

Section A

Answer all questions.

Each question carries 1 mark.

1. State the sufficient condition for the convergence of sequence of approximations $x_{n+1} = \phi(x_n)$ in iteration method.
2. Evaluate $\Delta^{10}(x^7)$.
3. Give Newton's divided difference formula.
4. Write the relation between forward and backward difference operators.
5. Form the table of differences of the function $f(x) = x^2$ for $x = 0, 1, 2, 3, 4, 5, 6$.
6. State Gauss' backward central difference formula.
7. Given a set of n -values of (x, y) , what is the formula for computing $\left[\frac{d^2y}{dx^2} \right]_{x_0}$.
8. What do you mean by numerical integration ?
9. State general formula for numerical integration.
10. What is partial pivoting and complete pivoting ?
11. State Predictor formula.
12. Write Runge-Kutta formula of fourth order to solve $\frac{dy}{dx} = f(x, y)$ with $y(x_0) = y_0$.

(12 × 1 = 12 marks)

Turn over

Section B

Answer at least **eight** questions.

Each question carries 6 marks.

All questions can be attended.

Overall Ceiling 48.

13. Find by iteration method, a real root of $2x - \log_{10} x = 7$.

14. Find the second divided difference of $f(x) = \frac{1}{x}$, using the points x_0, x_1, x_2 .

15. If D stands for the differential operator $\frac{d}{dx}$, prove that $D = \frac{1}{h} \left[\Delta - \frac{1}{2} \Delta^2 + \frac{1}{3} \Delta^3 - \dots \right]$.

16. Find the missing values in the following table :

x		45	50	55	60	65
y	:	3.0	—	2.0	—	— 2.4

17. The following table gives the population of a town during the last six censuses. Estimate, using Newton's interpolation formula, the increase in the population during the period 1946 to 1948 :

Year	:	1911	1921	1931	1941	1951	1961
Population (in thousands)		12	15	20	27	39	52

18. If $y_1 = 4, y_3 = 12, y_4 = 19$ and $y_x = 7$, find x .

19. The following data represents e^{-x} .

x	:	-1	-0.5	0	0.5	1
$f(x)$	:	2.7183	1.6487	1	0.6065	0.3679

Obtain approximate value of $f''(-1)$.

20. A curve is given by the points of the table given below :

x	:	0	0.5	1.0	1.5	2.0	2.5	3.0	3.5	4.0
y	:	23	19	14	11	12.5	16	19	20	20

Apply Simpson's rule to find the area bounded by the curve, the x -axis and the extreme ordinates.

21. Evaluate $\int_1^3 \frac{1}{x} dx$ by Simpson's rule with 8 strips. Determine the error by direct integration.
22. Decompose the matrix $\begin{bmatrix} 1 & 3 & 8 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{bmatrix}$ in the form LU.
23. Apply Gauss-Jordan method to solve the equations :
 $x + y + z = 9$; $2x - 3y + 4z = 13$; $3x + 4y + 5z = 40$.
24. Solve the system of equations $10x + 2y + z = 9$; $2x + 20y - 2z = -44$; $-2x + 3y + 10z = 22$ by Gauss-Seidel iteration method.
25. Given $\frac{dy}{dx} - 1 = xy$ and $y(0) = 1$, obtain the Taylor series for $y(x)$ and compute $y(0.1)$ correct to four decimal places.
26. Find the largest eigen value of the matrix $\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 4 \end{bmatrix}$.

(8 × 6 = 48 marks)

Section C

Answer at least five questions.

Each question carries 9 marks.

All questions can be attended.

Overall Ceiling 45.

27. Using Ramanujan's method find a real root of the equation $e^{-x} = x$.
28. Find an interval of unit length which contains the smallest negative root in magnitude of the equation $2x^3 + 3x^2 + 2x + 5 = 0$. Using the end points of this interval as initial approximation perform four iterations of the Regula-Falsi method.
29. Using Newton's forward difference formula, find the sum $S_n = 1^3 + 2^3 + 3^3 + \dots + n^3$.

Turn over

30. Interpolate by means of Gauss's backward formula the sales of a concern for the year 1976 given that :

Year	:	1940	1950	1960	1970	1980	1990
Sales (in Lakhs)	:	17	20	27	32	36	38

31. A function $f(x)$ representing the following data values has a minimum in the interval (0.5, 0.8). Find this point of minimum and the minimum value :

x	:	0.5	0.6	0.7	0.8
$f(x)$		2.3256	1.4632	0.9842	1.3282

32. Estimate the annual rate of cloth sales of 1935 from the following data :

Year	:	1920	1925	1930	1940
Sales of Cloth (in lakhs of metres)		250	285	328	444

33. Apply Lagrange's method to find the value of x when $f(x) = 15$ from the given data :

x	:	5	6	9	11
$f(x)$		12	13	14	16

34. Find the inverse of the matrix $\begin{bmatrix} 1 & -1 & 1 \\ 1 & -2 & 4 \\ 1 & 2 & 2 \end{bmatrix}$ using Gauss elimination method.

35. Solve, by Euler's method, the equation $\frac{dy}{dx} = x + y, y(0) = 0$. Choose $h = 0.2$ and compute $y(0.4)$ and $y(0.6)$.

(5 × 9 = 45 marks)

Section D

Answer any one question.

The question carries 15 marks.

36. (a) A rod is rotating in a plane about one of its ends. If the following table gives the angle θ radians which the rod has turned for different values of time t seconds, find its angular velocity and angular acceleration when $t = 0.7$ seconds :

t seconds	:	0.0	0.2	0.4	0.6	0.8	1.0
θ radians	:	0.0	0.12	0.48	1.10	2.0	3.20

- (b) Prove the following :

(i) $u_x = u_{x-1} + \Delta u_{x-2} + \Delta^2 u_{x-3} + \dots \Delta^{n-1} u_{x-n} + \Delta^n u_{x-n-1}$.

(ii) $\Delta \nabla y_k = \nabla \Delta y_k = \delta^2 y_k$.

(iii) $\Delta \left(\frac{1}{y_k} \right) = -\Delta y_k / (y_k y_{k+1})$.

37. Find the LU decomposition of the matrix

$$\begin{bmatrix} 1 & 0 & 5 & 2 \\ -1 & 4 & 1 & 0 \\ 3 & 0 & 4 & 1 \\ -2 & 1 & 1 & 3 \end{bmatrix}.$$

38. (a) Using Milne's method, obtain the solution of the differential equation

$$\frac{dy}{dx} = \frac{1}{x+y}, y(0) = 2 \text{ at } x = 0.8, \text{ given that } y(0.2) = 2.0933, y(0.4) = 2.1755, y(0.6) = 2.2493.$$

- (b) Determine the largest eigen value and corresponding eigen vector of the matrix $\begin{bmatrix} 1 & 6 & 1 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$.

(1 × 15 = 15 marks)